

Work Breakdown Structure (WBS)

Project Name: ConnectHub – Student Networking

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Level 1: Project - ConnectHub

1.0 Project Planning & Setup

- 1.1 Define scope and goals
 - 1.2 Create GitHub repository
 - 1.3 Finalize tech stack
 - 1.4 UI/UX Wireframe preparation
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2.0 Frontend Development (Next.js)

- 2.1 Initialize Next.js environment
 - 2.2 Implement login/signup using Clerk
 - 2.3 Create user profile pages
 - 2.4 Develop home feed (posts/projects)
 - 2.5 Build search and explore functionalities
 - 2.6 Implement notifications and responsiveness
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3.0 Backend Development (Node.js & Express.js)

- 3.1 Setup Express.js server
- 3.2 Design RESTful APIs for user, post, connection
- 3.3 Integrate MongoDB for database operations
- 3.4 Implement middleware, error handling, validation

4.0 Database Design (MongoDB)

- 4.1 User schema (profile, skills, links)
 - 4.2 Post schema (title, description, file links)
 - 4.3 Follower/Connection schema
 - 4.4 Resume/Research Uploads schema
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5.0 Media Integration (Cloudinary)

- 5.1 Cloudinary setup
 - 5.2 Upload handler for profile pics, resumes, projects
 - 5.3 Store URLs in MongoDB
 - 5.4 Validate and secure media uploads
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6.0 Authentication (Clerk)

- 6.1 Clerk configuration
 - 6.2 JWT & session integration
 - 6.3 Page protection (authenticated routing)
 - 6.4 Clerk profile data linking
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7.0 Testing

- 7.1 Unit testing of frontend components
- 7.2 Postman testing for API routes
- 7.3 Functional testing (signup, upload, connect)
- 7.4 UI/UX bug fixes